
A Robot Policy Designed as a Weight-Space Artifact Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and causal intervention

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Abstract

Tensor decomposition rewrites a multidimensional table as a small set of low-rank factors. In a neural network, those factors can expose which combinations of inputs and outputs are encoded by the weights. Conventional nonlinear operations make such a table difficult to obtain directly. We introduce χ -VLA, a 20.1M-parameter vision-language-action policy containing only bilinear, linear, and rational learned operations. Its checkpoint therefore specifies an explicit rational tensor program rather than only an opaque executable function. A mechanical audit and local reconstructions verify that claim. We derive an exact weight-based decomposition of each softmax-free attention layer and reconstruct all 12 modules in three checkpoints with worst relative error 2.56×10^{-7} . An $O(d^2)$ reduced Gram avoids materializing the coefficient tensor. All 36 primary block-head tests favor the rank-128 weight-derived subspace over fixed random controls. The full appendix screen finds the effect in 83/96 pairs and reveals depth-specific negative controls. Across three checkpoints, a causally ranked 64-dimensional visual subspace succeeds in 29/60 matched trials, compared with 47/60 for the full representation and 0/60 for random subspaces. The causal effect replicates, but near-lossless preservation does not. On disjoint cache samples, the selected basis has at least $9.9\times$ lower reconstruction error than the median of five random projectors in every action group and checkpoint. The artifact is behaviorally nontrivial, reaching $93.7\% \pm 0.8\%$ on LIBERO-Object. The first matched conventional twin differs by only 0.006% in parameters. Complete same-hardware replications find a 4.4-point rational deficit on A30 and 5.8 points on A6000. Compilation reduces the latency ratio from $3.49\text{--}5.33\times$ eager to $1.73\text{--}1.74\times$ compiled. A fifteen-configuration direct coefficient-edit sweep yields no candidate that passes both selectivity gates. We therefore establish a capable model artifact with exact layerwise structure, not yet a semantic, compute-matched, or selectively editable interface.

1 Introduction

Neural checkpoints are usually treated as opaque containers for functions. We can run them and collect their activations, but the weights alone rarely provide a readable map from input features to outputs. A post-hoc probe may reveal that an activation correlates with behavior, yet it does not show how the weights construct that feature or whether editing it will have a selective causal effect.

A tensor is a multidimensional table, and tensor decomposition is the analogue of low-rank matrix factorization for such tables. When a network uses only additions, multiplications, and ratios of polynomials, its weights can be expanded into coefficient tensors whose entries record input interactions. Decomposition ranks simpler factors inside those coefficients. The intended benefit is an

inspectable model artifact: candidate mechanisms come from the trained parameters, then activation data and interventions test whether they matter. Softmax, ordinary activations, and square-root normalization obstruct this direct conversion in conventional networks. Bilinear language and vision models suggest that changing the primitives can recover it [3, 4]. Whether the resulting artifact can still control a robot is unresolved.

We study two linked questions:

Can the learned weights of a specialist VLA specify an explicit rational tensor program, and can that artifact expose exact structure with a measurable causal consequence?

Our contributions are:

1. We construct and mechanically certify a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We derive exact weight-based structure through each softmax-free attention layer and an $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
3. We test the artifact at behavioral scale. Exact subspaces beat fixed random controls, and a causally ranked visual subspace consistently beats random controls across three checkpoints while remaining materially below the full policy.
4. We establish that the analyzed artifact is a capable policy, reaching $93.7\% \pm 0.8\%$ over three LIBERO-Object seeds, while reporting a language shortcut and a failed coefficient-surgery gate that bound the current interpretation. A same-skeleton conventional twin differs by only 0.006% in parameters. Two complete same-hardware replications place the seed-0 capability cost between 4.4 and 5.8 points.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

76 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 77 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

78 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 79 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 80 rescaling without leaving the rational tensor representation.

81 2.2 Architecture and training

82 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 83 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 84 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 85 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 86 continuous action dimensions. The complete model has 20,137,352 learned parameters.

87 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 88 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 89 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 90 that verify every requested state was loaded.

91 2.3 Mechanical certificate on the trained checkpoint

92 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 93 point at runtime and reconstruct representative operations from saved weights.

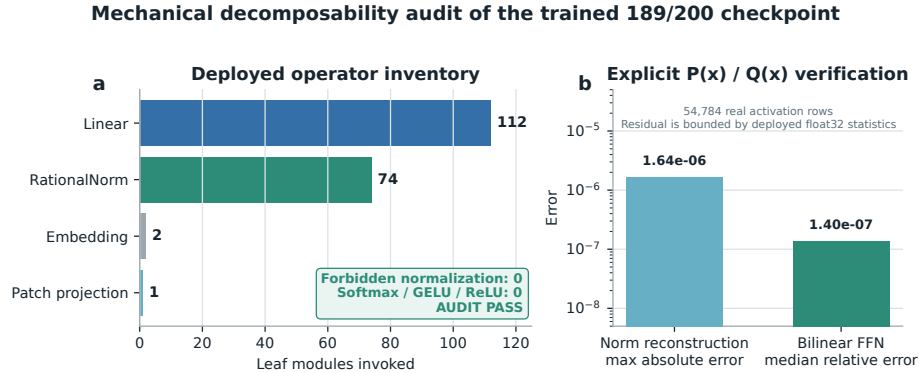


Figure 1: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

94 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 95 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 96 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does
 97 not materialize one compact polynomial for the full eight-layer transformer.

98 3 The artifact is a capable closed-loop policy

99 3.1 Evaluation protocol

100 We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
Same-skeleton conventional ViT	20.1M	1	89.8%
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

The same-skeleton conventional twin replaces the rational blocks with softmax attention, SwiGLU, and RMSNorm while holding the data, updates, seed, decoder, and 500 canonical trials fixed. On A30, the conventional twin succeeds in 448/500 trials and the rational checkpoint in 426/500. On A6000, they reach 451/500 and 422/500. Their parameter counts differ by only 0.006%. The rational deficit is 4.4 and 5.8 points, and task 3 accounts for 21 lost successes on each platform. Targeted replications reverse the task-3 gap: χ -VLA seeds 1 and 2 each reach 41/50, compared with 37/50 and 33/50 for their conventional counterparts. The seed-0 task gap is therefore not directionally consistent across seeds. Full-suite seed 1 reaches 404/500 for χ -VLA and 45/500 for the conventional checkpoint, but different trainer provenance prevents a conversion-cost interpretation. The collapsed conventional seeds fit the training cache with normalized MSE of 9.51×10^{-4} and 1.19×10^{-3} , which rules out gross checkpoint corruption but not closed-loop covariate shift. The same χ -VLA checkpoint reaches 423/500 on A40, between its A30 and A6000 results. On one A30, eager latency is 38.05 versus 7.14 ms. Compilation reduces it to 1.52 versus 0.88 ms, a $1.74\times$ gap. Rational training is $3.12\times$ slower at batch 128 and uses $1.63\times$ the peak allocated memory. A6000 profiling confirms a $1.73\times$ compiled ratio.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. Seed 0 misses the predeclared three-point noninferiority margin on both GPU classes. Conventional seeds 1 and 2 completed at only 9.0% and 8.2%, despite low final-minibatch losses and excellent training-cache fit. This exposes severe closed-loop seed sensitivity rather than a stable across-seed conversion cost. The weak checkpoints were trained natively on Athena, while the strong seed-0 checkpoint is a legacy artifact. A fully native matrix, matched χ -VLA evaluations, and partial-conversion controls remain necessary.

138 4 Exact weight-derived structure through attention

139 4.1 Exact layerwise construction

140 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 141 a coefficient table whose entries measure how combinations of query, key, and value coordinates
 142 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
 143 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
 144 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

145 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 146 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 147 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 148 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 149 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
 150 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
 151 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation's
 152 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

153 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 154 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
 155 rank important directions without constructing the full table. It matches brute-force materialization to
 156 2.13×10^{-14} at toy scale.

157 4.2 Trained policy result

158 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
 159 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
 160 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
 161 activations are used only afterward to score how well it preserves the layer's computation.

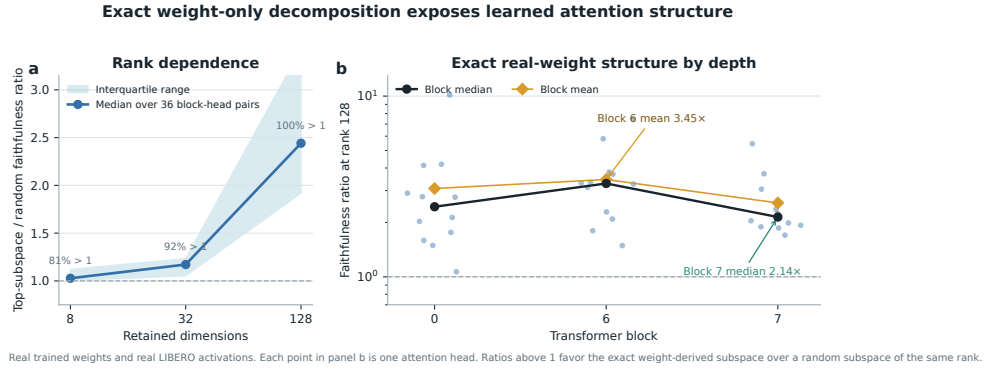


Figure 2: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

162 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 163 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 164 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 165 0, 6, and 7.

166 This extends exact weight-only construction through individual attention layers. It is not an exact
 167 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 168 rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation. Retaining it means zeroing all other directions, then running the actual policy. We compare selected and random equal-width subspaces. Because the ranking uses cached training demonstrations and downstream action gradients, offline reconstruction remains in-cache. Its evaluation samples are disjoint from the samples that estimate the ranking. Closed-loop intervention uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result.

An earlier one-checkpoint screen found in-cache random-to-selected reconstruction ratios of $4.8\times$, $4.6\times$, and $41.8\times$ for translation, rotation, and gripper, followed by 19/20, 18/20, and 0/20 closed-loop successes for full, selected, and random conditions. The three-checkpoint replication reaches 47/60, 29/60, and 0/60. Selected versus random has exact paired sign $p < 0.008$ at every checkpoint, but selected remains 30 points below full. The earlier near-lossless result does not replicate. Using 1,024 discovery samples and 1,024 disjoint evaluation samples per checkpoint, the selected basis has $16.2\text{--}32.3\times$ lower translation error, $9.9\text{--}15.0\times$ lower rotation error, and $15.4\text{--}30.9\times$ lower gripper error than the median of five equal-rank random projectors. This repairs sample reuse but does not turn the training-cache diagnostic into held-out task generalization. The exact layerwise attention method in Section 4 must still be connected to the same full-suite closed-loop intervention.

5.2 Counterfactual instruction test

High success does not guarantee robust language use. A one-step screen responds to a renamed visible object in only 15.0% of 80 states. We therefore run 100 paired 80-step rollouts for each of three checkpoints. Relative end-effector preference for the renamed object increases by 0.175, 0.168, and 0.189 m, with all trial-bootstrap 95% intervals above zero. Motion shifts toward that object in 84%, 86%, and 88% of pairs, yet only 33%, 28%, and 28% end closer to it. Appendix Figure 4 shows the replication. The seed-0 conventional twin has a 0.239 m shift, 92% directional response, and the same 33% endpoint rate as rational seed 0. Language response is therefore not unique to tensor decomposability. This is a reach-phase proximity outcome, not counterfactual task success.

The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target, so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy range.

This mixed result narrows the VLA claim. Language changes the trajectory reproducibly, but scene priors often prevent complete rerouting. The model is a capable language-conditioned specialist, not yet a robust instruction-grounded policy. LIBERO-CF independently documents the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively repair that shortcut.

6 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

218 **Limitations and conclusion.** The same-skeleton seed-0 control has a 4.4 to 5.8 point capability
219 gap and a residual compiled-runtime tax. Two additional conventional seeds collapse below 10%,
220 so training stability remains unresolved. The benchmark is one specialist suite, counterfactual
221 instructions do not fully reroute the policy, and exact decomposition is layerwise rather than end
222 to end. The closed-loop subspace test replicates across three checkpoints but loses 18 of the full
223 policy’s 47 successes, while a fifteen-configuration discovery sweep yields no coefficient edit that
224 passes both selectivity gates. The defensible result is therefore narrow: rational restrictions do not
225 prevent competitive LIBERO-Object control, and they provide a precise weight-derived analysis
226 surface. Broader grounding, multi-seed controls, optimized execution, and validated coefficient edits
227 remain necessary.

228 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
229 certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are
230 unavailable.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

262 A.2 Matched architecture capability and systems cost

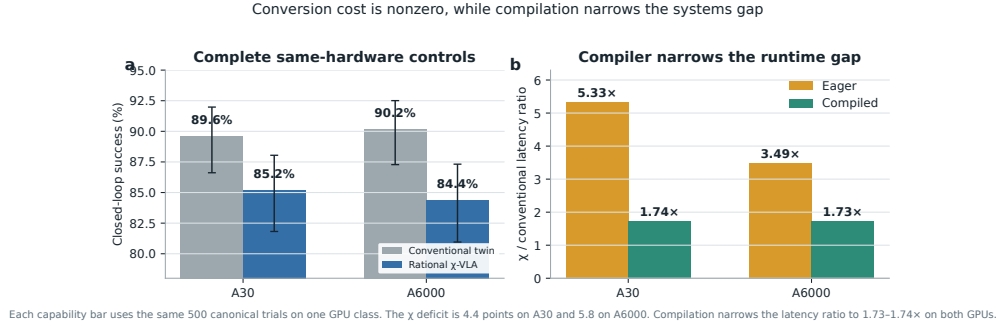


Figure 3: **Conversion cost is modest but nonzero, and compilation narrows the systems gap.** Complete same-hardware evaluations give 89.6% versus 85.2% on A30 and 90.2% versus 84.4% on A6000, conventional then rational. Parameters differ by 0.006%. Compilation narrows the batch-one latency ratio to 1.73–1.74 $\times$ on both GPUs.

263 A.3 Three-checkpoint counterfactual grounding

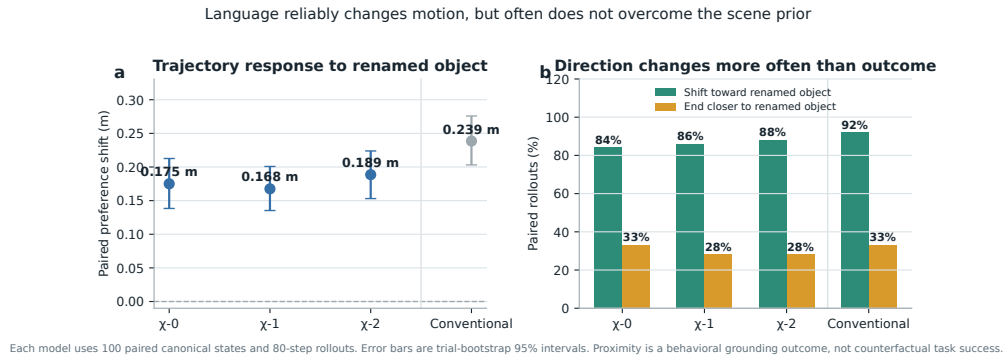


Figure 4: **Renaming the target changes motion reliably but does not fully override the scene prior.** Each model is evaluated on 100 paired canonical states for 80 control steps. The left panel reports the change in end-effector preference for the newly named object, with 95% bootstrap intervals. The right panel separates directional response from the stricter condition of ending closer to that object. These are proximity tests, not counterfactual task-success measurements. The conventional control shows that ordinary language response is not unique to tensor decomposability.

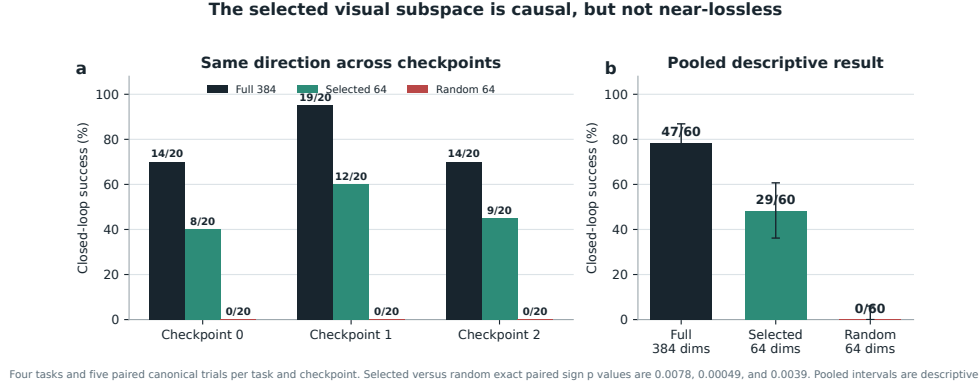


Figure 5: **The selected visual subspace has a replicated causal effect, but it is not near-lossless.** Each checkpoint uses four tasks and five paired canonical trials per task. Selected subspaces beat random controls in all three checkpoints. The pooled interval is descriptive because trials within a checkpoint share learned weights.

264 A.4 Per-task ensemble behavior

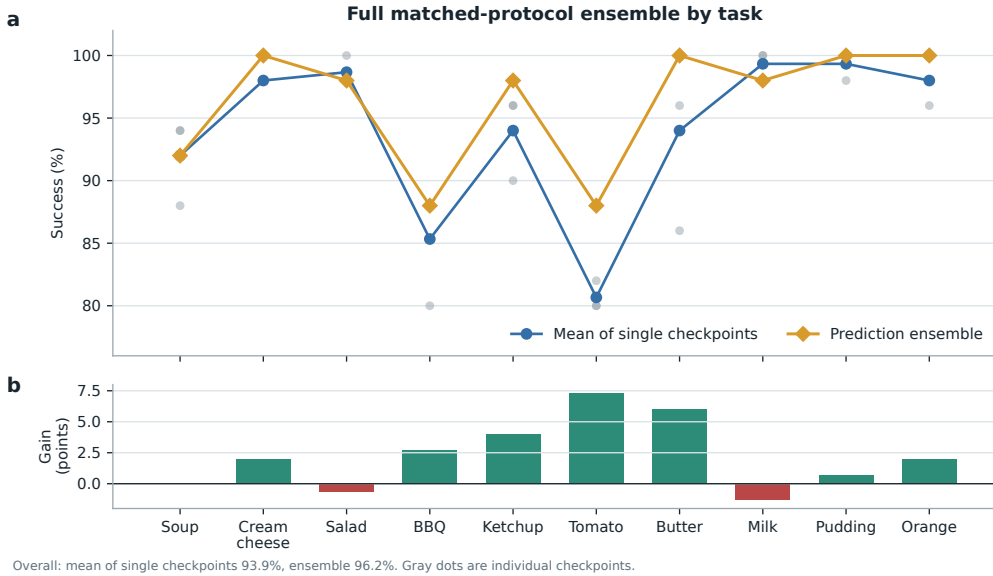


Figure 6: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

265 A.5 Product-routing multimodal head

266 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

267 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 268 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$

and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head. Product routing remains a construction result until it wins a controlled genuinely multimodal task.

A.6 Development interventions

Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points. A reward audit shows that accumulated step cost makes every successful trajectory have negative return. After removing that term and repairing the terminal boundary, one pilot improves by two points. These are development diagnostics, not validated paper contributions.

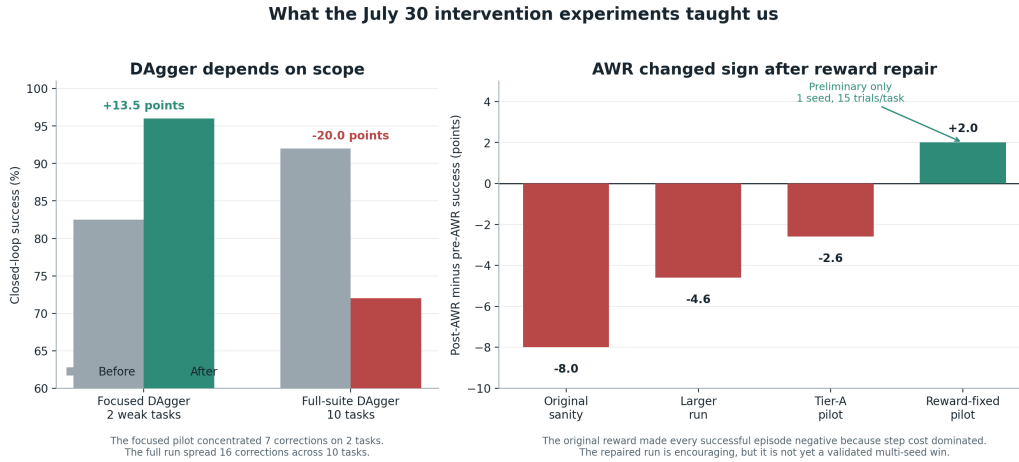


Figure 7: Development interventions reveal scope and reward sensitivity. Focused DAgger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

A.7 Expanded attention screen and first direct edit

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20, compared with 17/20 for both random controls, which is the wrong direction for the necessity prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and value column edits.

A subsequent discovery sweep evaluates blocks 0, 2, 4, 6, and 7 at ranks 64, 128, and 256. None of the fifteen configurations passes both prespecified 15-point gates. The best configuration, block 0 at rank 128, has a 10-point retention advantage and a 40-point removal advantage over Frobenius-matched random edits. It therefore does not advance to held-out confirmation. The selected keep and removal operations change 56.4% and 82.6% of the affected query, key, and value weights in relative Frobenius norm. This is a global functional ablation, not a small semantic edit.

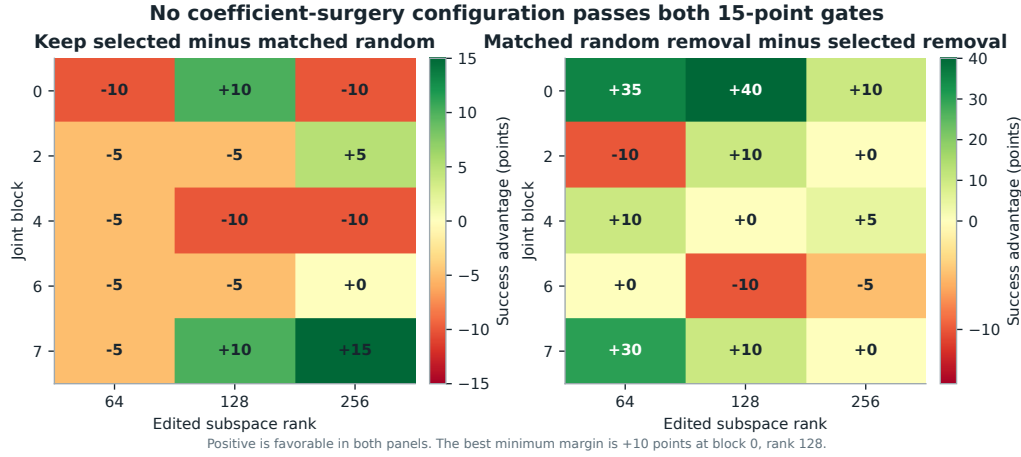


Figure 8: **The complete coefficient-surgery discovery sweep is negative.** Positive values favor the selected weight-derived subspace. No block-rank pair reaches the prespecified 15-point margin in both retention and removal.

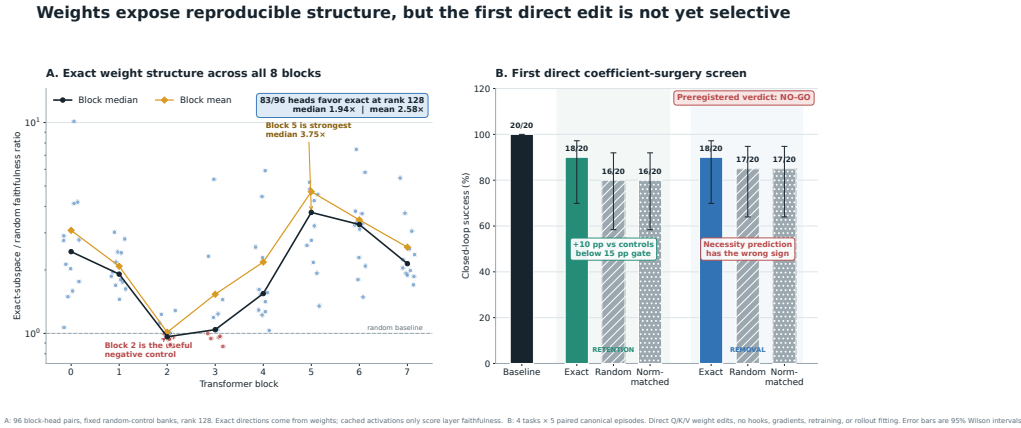


Figure 9: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

296 A.8 Numerical scope of rational conversion

297 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 298 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 299 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 300 establish practical numerical safety under distribution shift.